

**NEPAL COLLEGE OF INFORMATION TECHNOLOGY**  
**Assessment Spring 2025**

Level: Bachelor  
 Programme: BE\_COM(M & D)\_IT(DAY)  
 Course: Applied Mathematics

Semester – Spring

Year: 2025  
 Full Marks : 100  
 Time: 3 hrs.

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

**Attempt all the questions.**

1. a. ✓ State and prove cauchy-Reimann equation as necessary conditions for function to be analytic. 5
- b. ✓ Define Harmonic function. show that  $u = \cos x \cosh y$  is harmonic and find corresponding analytic function. 5
- c. ✓ State and prove cauchy –integral formula 5
2. a. ✓ Solve:  $\oint_C \frac{1}{z^2-1} dz$ , the counter clockwise around the circle  $|z-1|=1$ . (chapter-1) 5
- b. Define Laurent series. Find the Laurent expansion for  $f(z) = \frac{1}{z-z^3}$  in the region given by  $1 < |z+1| < 2$  5
- c. ✓ Define Cauchy –Residue theorem. By using Cauchy residue theorem evaluate the integration  $\oint_C \frac{z-23}{z^2-4z-5} dz$ , where  $C$  is the circle  $|z|=6$  5
3. a. ✓ Define bilinear transformation as well as fixed point. Find the bilinear transformation which makes the point  $z=1, i, -i$  onto  $w=i, 0, -i$ . Also find the image of unit circle  $|z|=1$  5
- b. State and prove first shifting theorem of Z transform. Using it to find the  $z$ -transform of  $Z(e^{-at} \cos wt)$ . 5
- c. State and prove Initial and final value problem of Z-transform. Find the  $z$ -transform of  $Z(a^n \sin bt)$ . 5
4. a. Prove that  $Z(y_{n+k}) = z^k (\bar{y} - y_0 - \frac{y_1}{z} - \dots - \frac{y_{k-1}}{z^{k-1}})$  5
- b. ✓ Solve the difference equation:  $y_{n+2} + 6y_{n+1} + 9y_n = 2^n$  with  $y_0 = y_1 = 0$  5
5. a. ✓ Using Fourier integral representation, show that 5
 
$$\int_0^\infty \frac{\cos xw + w \sin xw}{1+w^2} dw = 0 \text{ if } x < 0$$

$$= \frac{\pi}{2} \text{ if } x = 0$$

$$= \pi e^{-x} \text{ if } x > 0$$
- b. ✓ Find the Fourier sine transform of 5
6. a.  $f(x) = e^{-x}$  for  $x > 0$ . Then prove that  $\int_0^\infty \frac{\sin mx}{1+x^2} dx = \frac{\pi}{2} e^{-m}$  for  $m > 0$ . 5
- Solve the following equation: 5
 
$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0 \quad \text{by the separation of variables.}$$

b.

A tightly stretched string with fixed end points  $x=0$  and  $x=L$  is initially in a position given by  $u(x,0)=\sin x^3 \left(\frac{\pi x}{L}\right)$ . If it is released from rest from this position. Find the deflection  $u(x,t)$ .

5

OR

Derive one dimensional Heat equation.

c.

Determine the solution of one dimensional heat equation  $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$  Subject to the boundary condition  $u(0,t)=u(L,t)=0$  and initial condition is  $u(x,0)=L$ , being the length of the bar

5

OR

Derive one dimensional Wave equation with required assumptions.

7

a.

When two dimensional Heat equation  $\frac{\partial u}{\partial t} = c^2 \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$  become Laplace equation. By the concept of solution of Laplace equation with rectangular boundaries, solve the problem:

1+7

A rectangular plate with insulated surface is 8 cm wide so long compared to its width that it may be considered infinite in length without introducing an appreciable error. If the temperature along the short edge  $y=0$  is given by

$$u(x,0)=100 \sin \frac{\pi x}{8}, \quad 0 < x < 8$$

while two long edges  $x=0$  and  $x=8$  as well as the other short edge are kept at  $0^\circ\text{C}$ .

Show that steady state temperature at any point of the plate is given by

$$u(x,y)=100e^{-\pi y/8} \sin \frac{\pi x}{8}$$

OR

Derive two dimensional heat equation with required assumptions.

b.

Derive polar form of Laplace equation.

7

OR

A plate with insulated surface has the shape of quadrant of a circle of radius 10 cm. The bounding radii  $\theta = 0$  and  $\theta = \frac{\pi}{2}$  are kept at  $0^\circ\text{C}$  and temperature along the circular quadrant is kept at  $100(\pi\theta - 2\theta^2)^\circ\text{C}$  for  $0 \leq \theta \leq \frac{\pi}{2}$



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AM assessment Qsn

Q1a)

Cauchy Reimann equation states that, "A function  $f(z)$  is said to be analytic in domain  $D$  if and only if the partial derivatives of  $u$  and  $v$  satisfy the two conditions  $u_x = v_y, u_y = -v_x$  at every point in  $D$ ".

Let  $f(z) = u(x, y) + i v(x, y)$  be a complex valued function.

$$\text{or } f(z) = u + i v \quad \text{--- (i)}$$

$$\text{Now, } f(z + \Delta z) = (u + \Delta u) + i(v + \Delta v) \quad \text{--- (ii)}$$

subtracting (i) from (ii),

$$f(z + \Delta z) - f(z) = \Delta u + i \Delta v$$

$$\text{or, } \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta u + i \Delta v}{\Delta z}$$

$$\text{or, } f'(z) = \lim_{\Delta z \rightarrow 0} \left( \frac{\Delta u}{\Delta z} + i \frac{\Delta v}{\Delta z} \right)$$

We know,  $z = x + iy$

$$\text{or, } \Delta z = \Delta x + i \Delta y$$

(i) In real axis,  $\Delta y = 0$  so,  $\Delta z = \Delta x$

$$f'(z) = \lim_{\Delta z \rightarrow 0} \left( \frac{\Delta u}{\Delta x} + i \frac{\Delta v}{\Delta x} \right)$$

$$= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

(ii) In Imaginary axis,

$$f'(z) = \lim_{\Delta z \rightarrow 0} \left( \frac{\Delta u}{i\Delta y} + i \frac{\Delta v}{i\Delta y} \right)$$

$$f'(z) = i \frac{\partial u}{\partial y} - \frac{\partial v}{\partial y}$$

Now, we can write,

$$\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} + i \left( \frac{\partial u}{\partial y} \right)$$

Hence,

$$u_x = v_y \quad \text{and} \quad v_x = -u_y$$

This proves CR equation.

Q1b)

A complex valued function  $f(x,y) = u(x,y) + iv(x,y)$  is said to be harmonic function if the second partial derivative of function exists i.e.  $(u_{xx}, u_{yy}, v_{xx}, v_{yy})$  and satisfies Laplace equation (i.e.  $u_{xx} + u_{yy} = 0$  and  $v_{xx} + v_{yy} = 0$ ).

Given,  $u = \cos x \cosh y$

$$\frac{\partial u}{\partial x} = -\sin x \cosh y, \quad \frac{\partial^2 u}{\partial x^2} = -\cosh x \cosh y$$

$$\frac{\partial u}{\partial y} = \cos x \sinh y, \quad \frac{\partial^2 u}{\partial y^2} = \cos x \cosh y$$

Here,

$$u_{xx} + u_{yy} = 0$$

$$\text{or, } -\cos x \cosh y + \cos x \cosh y = 0$$

Hence, it is harmonic function.

By CR equation, we know,

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

Hence,  $v_y = -\sin x \cosh y$

Integrating with respect to  $y$ , we get,

$$v = -\sin x \cdot \sinh y + h(x)$$

where  $h(x)$  is function of  $x$  only

Differentiating w.r.t.  $x$ ,

$$v_x = -\cos x \cdot \sinh y + h'(x)$$

$$\text{Or, } -u_y = -\cos x \cdot \sinh y + h'(x)$$

$$\text{Or, } -\cos x \sinh y = -\cos x \sinh y + h'(x)$$

$$\therefore h'(x) = 0$$

Integrating, we get,

$$h'(x) = c$$

Hence,  $v = -\sin x \cdot \sinh y + c$  is harmonic conjugate of  $u$ .

Hence, required analytic function is

$$f(z) = u + iv$$

$$= (\cos x \cosh y) + i(-\sin x \cdot \sinh y + c) \quad \underline{\underline{Ans}}$$

Q1c)

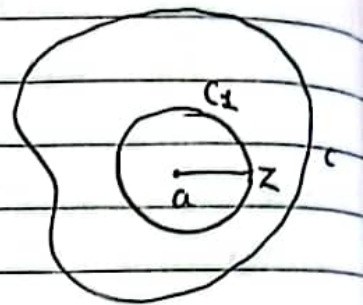
Cauchy Integral Formula:

Let  $f(z)$  is analytic within and on a closed curve  $c$ . Let  $a$  be any singular point inside  $c$  where  $f(z)$  is not analytic, then Cauchy integral formula states that,

$$f(a) = \frac{1}{2\pi i} \oint_c \frac{f(z)}{(z-a)} dz$$

Proof:

We have,  $f(z)$  which is analytic around a closed curve  $C$  except at point 'a'. Construct a circle  $C_1$  with center at  $a$  and small radius ( $r \rightarrow 0$ ).



Assume a point  $z$  on  $C_1$ . Now,

$$|z-a| = r$$

$$\text{or, } z-a = re^{i\theta}$$

$$\text{or, } z = a + re^{i\theta}$$

$$\text{or, } dz = rie^{i\theta} d\theta$$

From Cauchy integral theorem for multiple connected regions,

$$\oint_C \frac{f(z)}{(z-a)} dz = \oint_{C_1} \frac{f(z)}{(z-a)} dz$$

$$= \oint_{C_1} \frac{f(a+re^{i\theta})}{(a+re^{i\theta}-a)} rie^{i\theta} d\theta$$

$$= \oint_{C_1} f(re^{i\theta}) \times \frac{1}{re^{i\theta}} \times rie^{i\theta} d\theta$$

$$= i \oint_{C_1} f(re^{i\theta}) d\theta$$

Taking  $r \rightarrow 0$ ,

$$= i \oint_{C_1} f(a) d\theta$$

$$= i \times f(a) \int_0^{2\pi} d\theta$$

$$\text{or, } \oint_C \frac{f(z)}{(z-a)} dz = i \times f(a) 2\pi$$

$$\therefore f(a) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-a)} dz \quad \text{proved.}$$

$$\int_C \frac{1}{z^2-1} dz$$

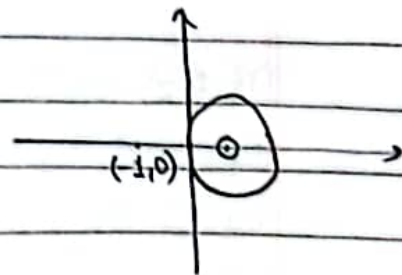
Here,  $f(z) = \frac{1}{z^2-1}$

To find singular point,  $z^2-1=0$

$$\text{or, } z^2 = 1$$

$$\text{or, } z = 1, -1$$

$$(1,0)$$



Given circle,  $|z-1|=1$

$$-1 \leq z-1 \leq 1$$

$$0 \leq z \leq 2$$

$$\text{Center} = \frac{2+0}{2} = 1 = (1,0)$$

$$\text{Radius} = 1$$

Here,  $z=1$  lies inside circle. so it is stationary point.

From Cauchy Residue theorem, Residue at  $z=1$  is,

$$R_1 = \lim_{z \rightarrow a} (z-a)f(z)$$

$$= \lim_{z \rightarrow 1} (z-1) \times \frac{1}{(z+1)(z-1)}$$

$$= \frac{1}{1+1} = \frac{1}{2}$$

Hence,  $\oint_C f(z) dz = 2\pi i (\text{sum of Residue})$

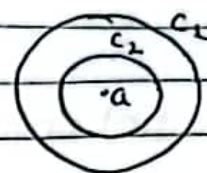
$$= 2\pi i \times \frac{1}{2}$$

$$\oint_C f(z) dz = \pi i$$

Ans

Q2b)

If  $f(z)$  be any analytic function in circle  $C_1$  and  $C_2$  and in region  $R$  bounded by  $C_1$  and  $C_2$ , then at any point  $z$  in region  $R$ ,  $f(z)$  can be written as series



in the form of positive and negative power of  $(z-a)$  in form

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-a)^n}$$

This is called Laurent series.

Given,  $f(z) = \frac{1}{z-z^3}$

$$= \frac{1}{z(1-z^2)}$$

$$f(z) = \frac{1}{z(1+z)(1-z)}$$

Given region is  $1 < |z+1| < 2$

Let,  $z+1 = u$

Or,  $z = u-1$

So,  $f(u) = \frac{1}{(u-1)(1+u-1)(1-u+1)}$

$$f(u) = \frac{1}{u(u-1)(2-u)} = \frac{A}{u} + \frac{B}{u-1} + \frac{C}{2-u}$$

Now, region is  $1 < |u| < 2$

Or,  $\frac{1}{u(u-1)(2-u)} = A(u-1)(2-u) + B(u)(2-u) + C(u)(u-1)$

Put  $u=1$ ,

$$1 = B \times 1 \times (2-1)$$

$$\therefore B = 1$$

Put  $u=2$ ,

$$1 = C \times 2 \times 1$$

$$\therefore C = \frac{1}{2}$$

Put  $u=0$ ,

$$1 = A \times (-1) \times (2)$$

$$\therefore A = -\frac{1}{2}$$

$$\text{So, } f(u) = \frac{-1}{2u} + \frac{1}{u-1} + \frac{1}{2(2-u)}$$

For the given region,  $1 < |u| < 2$ 

$$f(u) = \frac{-1}{2u} + \frac{1}{u \left(1 - \frac{1}{u}\right)} + \frac{1}{2u \left(\frac{2}{u} - 1\right)}$$

$$= \frac{-1}{2u} + \frac{1}{u} \times \left(1 - \frac{1}{u}\right)^{-1} + \frac{1}{2u} \left(\frac{1-2}{u}\right)^{-1}$$

$$= \frac{-1}{2u} + \frac{2}{2u} \sum_{n=0}^{\infty} \left(\frac{1}{u}\right)^n - \frac{1}{2u} \sum_{n=0}^{\infty} \left(\frac{2}{u}\right)^n$$

$$f(u) = \frac{1}{2u} \left[ -1 + 2 \sum_{n=0}^{\infty} \left(\frac{1}{u}\right)^n - \sum_{n=0}^{\infty} \left(\frac{2}{u}\right)^n \right]$$

$$\therefore f(z) = \frac{1}{2z+2} \left[ -1 + 2 \sum_{n=0}^{\infty} \left(\frac{1}{z+1}\right)^n - \sum_{n=0}^{\infty} \left(\frac{2}{z+1}\right)^n \right]$$

is the required Laurent series. Ans
(Q2c)

Cauchy Residue theorem states that, If  $f(z)$  is analytic function inside on a curve at any point except at number of poles (singular points) then  $f(z)$  can be written as,

$$\oint f(z) dz = 2\pi i \times (\text{Sum of Residue})$$

$$\text{Here, } f(z) = \frac{z-23}{z^2-4z-5}$$

$$\text{For singular point, } z^2-4z-5=0$$

$$\text{or, } z^2-5z+z-5=0$$

$$\text{or, } z(z-5)+1(z-5)=0$$

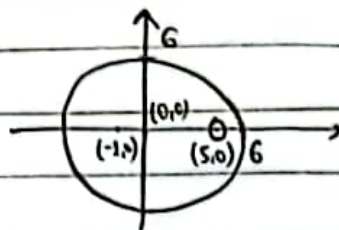
$$\text{or, } (z-5)(z+1)=0$$

 $z=5$  of order 1,  $z=-1$  of order 1

Given circle,  $|z| = 6$

Center = 0

radius = 6



Thus, both points lie inside circle.

For  $z = 5$ ,

$$R_1 = \lim_{z \rightarrow a} (z-a)f(z)$$

$$= \lim_{z \rightarrow 5} (z-5) \times \frac{(z-23)}{(z-5)(z+1)}$$

$$= \frac{5-23}{5+1} \quad \therefore R_1 = -3$$

For  $z = -1$ ,

$$R_2 = \lim_{z \rightarrow -1} (z+1)f(z)$$

$$= \lim_{z \rightarrow -1} (z+1) \frac{(z-23)}{(z+1)(z-5)}$$

$$= \frac{-1-23}{-1-5} \quad \therefore R_2 = 4$$

Hence, From Cauchy Residue theorem,

$$\oint f(z) dz = 2\pi i (\text{sum of residue})$$

$$= 2\pi i \times (-3+4)$$

$$\therefore \oint f(z) dz = 2\pi i$$

Ans

2a)

A transformation of the form  $f(z) = w = \frac{az+b}{cz+d}$  where  $ad-bc \neq 0$

is said to be bilinear transformation.

A fixed point in a bilinear transformation is a point in the complex plane that is mapped to itself i.e.  $w = z$ .

If  $w = \frac{az+b}{cz+d}$ , in fixed point,  $w = z$ . So,  $z = \frac{az+b}{cz+d}$

Given,  $z_1 = 1$

$w_1 = i$

$z_2 = i$

$w_2 = 0$

$z_3 = -i$

$w_3 = -i$

By using cross ratio,

$$\frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)} = \frac{(w-w_1)(w_2-w_3)}{(w-w_3)(w_2-w_1)}$$

$$\text{or, } \frac{(z-1)(i+i)}{(z+i)(i-1)} = \frac{(w-i)(0+i)}{(w+i)(-i-0)}$$

$$\text{or, } \frac{(z-1)(2i)}{(z+i)(i-1)} = \frac{(w-i)}{(-w-i)}$$

$$\text{or, } (2iz-2i)(-w-i) = (w-i)(z+i)(i-1)$$

$$\text{or, } -2i wz + 2i^2 = (wz + wi - iz - i^2)(i-1)$$

$$\text{or, } -2i wz - 2 = iwz - wz + i^2 w - iw - i^2 z + iz - i^3 + i^2$$

$$\text{or, } -2i wz - 2 = iwz - wz - w - iw + z + iz + i - 1$$

$$\text{or, } 3i wz + 1 - wz - w - iw + z + iz + i = 0$$

$$\text{or, } z(3iw - w + 1 + i) = w + iw - i - 1$$

or,  ~~$z(3i-1-i)$~~

$$w(3i-1-i) =$$

$$-1-z-iz-i$$

$$\therefore z = \frac{w+iw-i-1}{3iw-w+1+i} = \frac{(1+i)w-(1+i)}{(3i-1)w+(1+i)}$$

is the required transformation.

$$\therefore w = \frac{1+z+iz+i}{z+1+i-3iz} = \frac{(1+i)z+(1+i)}{(1-3i)z+(1+i)}$$

The image of unit circle  $z=1$  is,

$$z = 1$$

$$\text{or, } \frac{w+iw-i-1}{3iw-w+1+i} = 1$$

$$\text{or, } w+iw-i-1 = 3iw-w+1+i$$

$$\text{or, } 2iw-2w+2i+2 = 0$$

$$\text{or, } 2iu+2i^2v-2u-2iv+2i+2=0$$

$$\text{or, } 2iu-2v-2u-2iv+2i+2=0$$

$$\text{or, } u(2i-2)-v(2i+2)+(2i+2)=0$$

$$\text{or, } u(i-1)-v(i+1)+(i+1)=0$$

$$\text{or, } v(i+1)-u(i-1)=(i+1) \quad \text{is the image of unit circle}$$

Q3b)

First shifting theorem of z transform states that,

$$\text{If } Z[f(t)] = f(z)$$

$$\text{then } Z[e^{-at} f(t)] = f(ze^{at})$$

Proof:-

From the definition of z-transform,

$$Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n} = F(z)$$

Now,

$$Z[e^{-at} f(t)] = \sum_{n=0}^{\infty} e^{-anT} f(nT) z^{-n}$$

$$= \sum_{n=0}^{\infty} (e^{+aT} z)^{-n} f(nT)$$

$$= F(ze^{aT}) \quad \text{proved}$$

To find,  $z(e^{-at} \cos wt)$

unit  
circle

3c)

### Initial value theorem:

If  $Z[f(t)] = F(z)$ , then  $f(0) = \lim_{z \rightarrow \infty} F(z) = \lim_{t \rightarrow 0} f(t)$

Proof: By definition of z-transform,

$$F(z) = \sum_{n=0}^{\infty} f(nT) z^{-n}$$

$$F(z) = f(0) + \frac{f(T)}{z} + \frac{f(2T)}{z^2} + \dots$$

Taking limit  $z \rightarrow \infty$  on both sides,

$$\lim_{z \rightarrow \infty} F(z) = \lim_{z \rightarrow \infty} \left[ f(0) + \frac{f(T)}{z} + \frac{f(2T)}{z^2} + \dots \right]$$

$$= f(0)$$

$$= \lim_{t \rightarrow 0} f(t)$$

### Final value theorem:

If  $Z[f(t)] = F(z)$ , then  $f(\infty) = \lim_{z \rightarrow 1} (z-1)F(z)$

Proof: By definition of z-transform,

$$F(z) = Z[f(t)] = \sum_{n=0}^{\infty} f(nT) z^{-n} \quad \text{--- (1)}$$

$$z[F(z) - f(0)] = Z[f(t+T)] = \sum_{n=0}^{\infty} f(nT+T) \cdot z^{-n} \quad \text{--- (2)}$$

subtracting (1) from (2),

$$zF(z) - zf(0) - F(z) = \sum_{n=0}^{\infty} [f(nT+T) - f(nT)] z^{-n}$$

$$\text{or, } (z-1)F(z) = zf(0) + \sum_{n=0}^{\infty} [f(nT+T) - f(nT)] z^{-n}$$

Taking limit  $z \rightarrow 1$  on both sides,

$$\lim_{z \rightarrow 1} (z-1) F(z) = \lim_{z \rightarrow 1} \left\{ z f(0) + \sum_{n=0}^{\infty} [f(n\tau + \tau) - f(n\tau)] \right\}$$

$$\therefore \lim_{z \rightarrow 1} (z-1) F(z) = f(0) + \lim_{k \rightarrow \infty} \sum_{n=0}^k [f(n\tau + \tau) - f(n\tau)]$$

$$= f(0) + \lim_{k \rightarrow \infty} [f(\tau) - f(0) + f(2\tau) - f(\tau) + f(3\tau) - f(2\tau) + \dots + f(k\tau) - f((k-1)\tau)]$$

$$\therefore \lim_{k \rightarrow \infty} (z-1) F(z) = f(0) + \lim_{k \rightarrow \infty} [-f(0) + f(k+1)\tau]$$

$$= \lim_{k \rightarrow \infty} f((k+1)\tau) = f(\infty) \quad \text{Hence proved.}$$

Here,  $z(a^n e^{ibt}) = [z(e^{ibt})]_{z \rightarrow \frac{z}{a}}$

$$= \left( \frac{z}{z - e^{ibt}} \right)_{z \rightarrow \frac{z}{a}}$$

$$= \left( \frac{\frac{z}{a}}{\frac{z}{a} - e^{ibt}} \right)$$

$$= \frac{z}{z - a e^{ibt}}$$

or,  $z(a^n \cos bt + b \sin bt) = \frac{z}{z - a(\cos b\tau + i \sin b\tau)}$

$$= \frac{z}{(z - a \cos b\tau) + i(a \sin b\tau)} \times \frac{(z - a \cos b\tau) + i(a \sin b\tau)}{(z - a \cos b\tau) + i(a \sin b\tau)}$$

$$= \frac{z[(z - a \cos b\tau) + i(a \sin b\tau)]}{(z - a \cos b\tau)^2 + (a \sin b\tau)^2}$$

$$= \frac{z(z - a \cos b\tau)}{z^2 - 2az \cos b\tau + a^2} + \frac{ia z \sin b\tau}{z^2 - 2az \cos b\tau + a^2}$$

Hence, we obtain,

$$z(a^n \sin bt) = \frac{az \sin b\tau}{z^2 - 2az \cos b\tau + a^2} \quad \text{Ans}$$

Q4a) To Prove,  $Z(y_{n+k}) = Z^k \left( \bar{y} - y_0 - \frac{y_1}{Z} - \dots - \frac{y_{k-1}}{Z^{k-1}} \right)$

Proof:

By definition, we have,

By definition, we have,

$$\begin{aligned} Z(y_{n+k}) &= \sum_{n=0}^{\infty} y_{n+k} Z^{-n} \\ &= \sum_{n=0}^{\infty} y_{n+k} Z^{-(n+k)} Z^k \\ &= Z^k \sum_{n=0}^{\infty} y_{n+k} Z^{-(n+k)} \end{aligned}$$

Putting  $m = n+k$ , when  $n=0$ ,  $m=k$   
when  $n=\infty$ ,  $m=\infty$

$$\begin{aligned} Z(y_{n+k}) &= Z^k \sum_{m=k}^{\infty} y_m Z^{-m} \\ &= Z^k \left[ \sum_{m=0}^{\infty} y_m Z^{-m} - \sum_{m=0}^{k-1} y_m Z^{-m} \right] \\ &= Z^k \left[ \bar{y} - y_0 - \frac{y_1}{Z} - \frac{y_2}{Z^2} - \dots - \frac{y_{k-1}}{Z^{k-1}} \right] \end{aligned}$$

where  $\bar{y} = Z(y_n)$  Proved.

Q4b)

$$\begin{aligned} Z(y_{n+2}) + 6Z(y_{n+1}) + 9Z(y_n) &= Z(2^n) \\ \text{or, } Z^2 \left( \bar{y} - y_0 - \frac{y_1}{Z} \right) + 6 \left( \bar{y} - y_0 \right) + 9\bar{y} &= \frac{Z}{Z-2} \end{aligned}$$

$$\text{or, } \bar{y} (Z^2 + 6Z + 9) = \frac{Z}{Z-2}$$

$$\therefore \bar{y} = \frac{Z}{(Z-2)(Z+3)^2} \quad \text{--- (1)}$$

$$= \frac{Az}{(z-2)} + \frac{Bz}{(z+3)} + \frac{Cz}{(z+3)^2}$$

On solving,  $A = \frac{1}{25}$ ,  $B = -\frac{1}{25}$ ,  $C = -\frac{1}{5}$

From eq<sup>n</sup> (i),

$$\bar{y} = \frac{1}{25} \left( \frac{z}{z-2} \right) - \frac{1}{25} \left( \frac{z}{z+3} \right) - \frac{1}{5} \left( \frac{z}{(z+3)^2} \right)$$

Hence,

$$y_n = \frac{1}{25} (2)^n - \frac{1}{25} (-3)^n - \frac{1}{5} n (-3)^n \quad \underline{\underline{Ans}}$$

Q5a)

Comparing with fourier integral, this problem is fourier integral of  $f(n)$  where,

$$f(n) = \begin{cases} 0 & \text{if } n < 0 \\ \pi/2 & \text{if } n = 0 \\ \pi e^{-n} & \text{if } n > 0 \end{cases}$$

$$\text{Hence, } f(n) = \int_0^{\infty} (A(\omega) \cos \omega n + B(\omega) \sin \omega n) d\omega$$

Now,

$$A(\omega) = \int_{-\infty}^{\infty} f(t) \cos \omega t dt$$

$$= \frac{1}{\pi} \left[ \int_{-\infty}^0 0 \cos \omega t dt + \int_0^{\infty} \frac{\pi}{2} \cos \omega t dt + \int_0^{\infty} \pi e^{-t} \cos \omega t dt \right]$$

$$= \frac{e^{-t}}{(-1)^2 \omega^2} [-\cos \omega t + \omega \sin \omega t]_0^{\infty}$$

$$= \frac{1}{1+\omega^2} \left[ \frac{-\cos \omega t}{e^t} + \frac{\omega \sin \omega t}{e^t} \right]_0^{\infty}$$

$$= \frac{1}{1+\omega^2}$$

$$B(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(t) \sin \omega t dt$$

$$= \frac{1}{\pi} \times \pi \int_0^{\infty} e^{-t} \sin \omega t dt$$

$$= \left[ \frac{e^{-t}}{(-1)^2 + \omega^2} [-\sin \omega t - \omega \cos \omega t] \right]_0^{\infty}$$

$$= \frac{\omega}{1 + \omega^2}$$

substituting,

$$f(\eta) = \int_0^{\infty} \frac{1}{1 + \omega^2} (\cos \eta \omega + \omega \frac{\sin \eta \omega}{1 + \omega^2}) d\omega$$

$$\therefore \int_0^{\infty} \frac{\cos \eta \omega + \omega \sin \eta \omega}{1 + \omega^2} d\omega = 0 \quad \text{if } \eta < 0$$

$$= \frac{\pi}{2} \quad \text{if } \eta = 0$$

$$= \pi e^{-\eta} \quad \text{if } \eta > 0$$

Q5b)

Given  $f(\eta) = e^{-\eta}$

$$F_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin \omega t dt$$

$$= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-t} \sin \omega t dt$$

$$= \sqrt{\frac{2}{\pi}} \times \left[ \frac{e^{-t}}{(-1)^2 + \omega^2} ((-1) \sin \omega t + \omega \cos \omega t) \right]_0^{\infty}$$

$$= \sqrt{\frac{2}{\pi}} \times \frac{1}{1 + \omega^2}$$

We know,  $f(\eta) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(\omega) \sin \omega \eta d\omega$

$$\text{or, } e^{-\eta} = \sqrt{\frac{2}{\pi}} \int_0^{\infty} F_s(\omega) \sin \omega \eta d\omega$$

Put  $\eta = m$

$$\text{or, } e^{-\eta} = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1+w^2} \sin w \eta dw$$

Put  $\eta = m, w = n,$

$$e^{-m} = \frac{2}{\pi} \int_0^{\infty} \frac{1}{1+n^2} \sin n m dn$$

$$\therefore \int_0^{\infty} \frac{1}{1+n^2} \sin n m dn = \frac{\pi}{2} e^{-m} \quad \underline{\underline{Ans}}$$

Q6a) 
$$\frac{\partial^2 z}{\partial n^2} - 2 \frac{\partial z}{\partial n} + \frac{\partial z}{\partial y} = 0$$

Let  $z(n, y) = X(n) Y(y)$  be the solution.

$$\text{Now, } \frac{\partial^2 z}{\partial n^2} = z_{nn} = X'' Y$$

$$\frac{\partial z}{\partial n} = z_n = X' Y$$

$$\frac{\partial z}{\partial y} = z_y = X Y'$$

Now,

$$X'' Y - 2 X' Y + X Y' = 0$$

$$\text{or, } \cancel{X'' Y}$$

$$\text{or, } (X'' - 2X') Y + X Y' = 0$$

$$\text{or, } (X'' - 2X') Y = -X Y'$$

$$\text{or, } \frac{X'' - 2X'}{X} = \frac{-Y'}{Y} = K (\text{constant}) \quad \text{--- (1)}$$

Here, From (1)

$$\frac{X'' - 2X'}{X} = K$$

$$\text{or, } X'' - 2X' - XK = 0$$

The auxiliary eq<sup>n</sup> is,  $m^2 - 2m - K = 0$

$$ax^2+bx+c=0$$

$$\text{or, } m^2-2m-k=0$$

$$m = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$$= \frac{+2 \pm \sqrt{4-4 \times 1 \times (-k)}}{2 \times 1}$$

$$= \frac{2 \pm \sqrt{4+4k}}{2}$$

$$= 1 \pm \sqrt{1+k}$$

$$\text{Hence, } m = 1 + \sqrt{1+k} \text{ and } m = 1 - \sqrt{1+k}$$

So, the solution is,

$$X(n) = C_1 e^{(1+\sqrt{1+k})n} + C_2 e^{(1-\sqrt{1+k})n}$$

From (i),

$$\frac{-y'}{y} = k$$

$$\text{or, } -y' - yk = 0$$

$$\text{or, } y' + yk = 0$$

The auxiliary eqn is  $m+k=0$

$$\therefore m = -k$$

$$\text{So, } y(y) = C_3 e^{-ky}$$

Hence,

$$\begin{aligned} z(n,y) &= X(n) \cdot y(y) \\ &= (C_1 e^{(1+\sqrt{1+k})n} + C_2 e^{(1-\sqrt{1+k})n}) \cdot C_3 e^{-ky} \end{aligned}$$

Ans

$x=0$  and  $x=L$

$$u(x,0) = \sin x^3 \left( \frac{\pi x}{L} \right) = f(x)$$

$$u(x,t) = ?$$

$$g(x) = \text{Initial velocity} = 0$$

We know solution of 1D wave eq<sup>n</sup> is,

$$u(x,t) = \sum_{n=1}^{\infty} (a_n \cos \lambda_n t + b_n \sin \lambda_n t) \sin \frac{n\pi x}{L}$$

$$\text{where, } a_n = \frac{2}{L} \int_0^L f(x) \cdot \sin \frac{n\pi x}{L} dx$$

$$b_n = \frac{2}{L \cdot \lambda_n} \int_0^L g(x) \cdot \sin \frac{n\pi x}{L} dx$$

$$\text{As } g(x) = 0, \text{ so, } b_n = 0$$

$$\text{Now, } a_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

$$= \frac{2}{L} \int_0^L \sin x^3 \left( \frac{\pi x}{L} \right) \cdot \sin \left( \frac{n\pi x}{L} \right) dx$$

$$= \frac{1}{L} \int_0^L 2 \sin \left( \frac{\pi x^4}{L} \right) \cdot \sin \left( \frac{n\pi x}{L} \right) dx$$

$$= \frac{1}{L} \int_0^L \left[ \cos \left( \frac{\pi x^4}{L} - \frac{n\pi x}{L} \right) - \cos \left( \frac{\pi x^4}{L} + \frac{n\pi x}{L} \right) \right] dx$$

$$= \frac{1}{L} \int_0^L \left[ \cos \frac{\pi}{L} (x^4 - nx) - \cos \frac{\pi}{L} (x^4 + nx) \right] dx$$

(10)  $\frac{1}{(x^2+1)^2}$

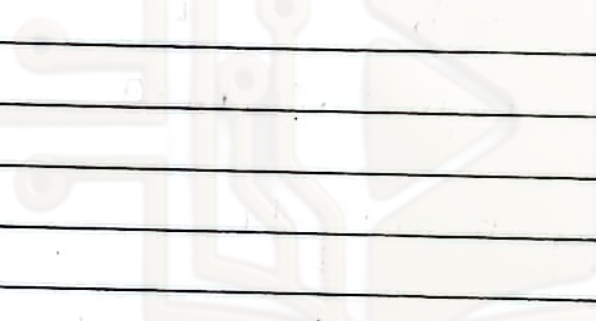
$\frac{1}{(x^2+1)^2}$

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$\frac{1}{(x^2+1)^2}$

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{--- (*)}$$

Boundary condition:  $u(0,t) = u(L,t) = 0$

Initial condition:  $u(x,0) = 1$

Let  $u(x,t) = X(x)T(t)$  be the sol<sup>n</sup> of eq<sup>n</sup> (\*),

$$\text{Now, } u_t = \frac{\partial u}{\partial t} = XT'$$

$$u_{xx} = \frac{\partial^2 u}{\partial x^2} = X''T$$

$$\text{Now, } XT' = c^2 X''T$$

$$\text{Or, } \frac{X''}{X} = \frac{T'}{Tc^2} = k \text{ (constant)}$$

$$\text{Now, } X'' - kX = 0$$

Auxiliary eq<sup>n</sup> is  $m^2 - k = 0$

$$\therefore m = \pm \sqrt{k}$$

$$\text{So, } X(x) = C_1 e^{kx} + C_2 e^{-kx}$$

$$\text{Then, } T' - Tc^2 k = 0$$

$$\text{Auxiliary eq<sup>n</sup> is } m + c^2 k = 0$$

$$\therefore m = -c^2 k$$

① If constant  $k$  is negative i.e.  $k = -p^2$ , Then, eq<sup>n</sup> we get is now,

$$X'' + p^2 X = 0$$

$$\text{A.E. is } m^2 + p^2 = 0$$

$$\text{Or, } m^2 = -p^2$$

$$\therefore m = 0 \pm ip$$

$$T' + Tc^2 p^2 = 0$$

$$\text{A.E. is } m + c^2 p^2 = 0$$

$$\text{Or, } m = -c^2 p^2$$

$$\text{Hence, } T(t) = C_3 e^{-c^2 p^2 t}$$

$$X(x) = (C_1 \cos px + C_2 \sin px) e^{0x}$$

$$= C_1 \cos px + C_2 \sin px$$

$$\text{If } t=0 \text{ then } \therefore u(x,t) = X(x) \cdot T(t)$$

$$u(x,t) = (C_1 \cos px + C_2 \sin px) C_3 e^{-c^2 p^2 t} \quad \text{--- ①}$$

If  $t \rightarrow \infty$ , then  $u(x,t) = e^{-\infty} \rightarrow 0$ . So, appropriate sol<sup>n</sup> of heat eq<sup>n</sup> is with constant negative

Using boundary condition  $u(0,t) = 0$  in (i),  
 $u(0,t) = (C_1 \cos 0 + C_2 \sin 0) \cdot C_3 e^{-C^2 P^2 t}$   
 $0 = C_1 C_3 e^{-C^2 P^2 t}$

As  $C_3 \neq 0$ , so,  $C_1 = 0$ .

Hence, in (i),

$$u(x,t) = C_2 \sin Px \cdot C_3 e^{-C^2 P^2 t} \quad \text{--- (ii)}$$

Using Next BC  $u(L,t) = 0$  in (ii),

$$u(L,t) = C_2 \sin PL \cdot C_3 e^{-C^2 P^2 t}$$

$$\text{Or, } 0 = C_2 \sin PL \cdot C_3 e^{-C^2 P^2 t}$$

$\rightarrow C_2, C_3$  can't be zero and for  $e^{-C^2 P^2 t}$  to be zero  $t \rightarrow \infty$ ,

$$\text{So, } \sin PL = 0$$

$$\sin PL = \sin n\pi$$

$$\therefore P = \frac{n\pi}{L}$$

Now,

$$u(x,t) = C_2 \sin\left(\frac{n\pi}{L}x\right) \cdot C_3 e^{-C^2 \left(\frac{n\pi}{L}\right)^2 t}$$

$$u(x,t) = b_n \sin\left(\frac{n\pi}{L}x\right) \cdot e^{-t\left(\frac{n\pi C}{L}\right)^2} \quad \text{where } b_n = C_2 \cdot C_3$$

The general solution is,

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) \cdot e^{-t\left(\frac{n\pi C}{L}\right)^2}$$

Using initial condition,

Q7a)

A two dimensional heat equation becomes Laplace equation at steady state condition.

At steady state condition,  $u$  is independent with time  $t$  i.e.  $\frac{\partial u}{\partial t} = 0$ .

Thus,  $u_{xx} + u_{yy} = 0$ , This is Laplace equation.

Given,  $L = 8 \text{ cm}$

$$u(x, 0) = 100 \sin \frac{\pi x}{8} = f(x)$$

The sol<sup>n</sup> of 2D heat eq<sup>n</sup> under given assumptions is given by,

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{8} \cdot e^{-\frac{n\pi y}{8}}$$

$$B_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{8} \int_0^8 f(x) \sin\left(\frac{n\pi x}{8}\right) dx$$

$$= \frac{2}{8} \int_0^8 100 \sin \frac{\pi x}{8} \cdot \sin \frac{n\pi x}{8} dx$$

$$= \frac{100}{8} \int_0^8 2 \sin \frac{\pi x}{8} \cdot \sin \frac{n\pi x}{8} dx$$

$$= \frac{100}{8} \int_0^8 \left( \cos\left(\frac{\pi x}{8} - \frac{n\pi x}{8}\right) - \cos\left(\frac{\pi x}{8} + \frac{n\pi x}{8}\right) \right) dx$$

$$= \frac{100}{8} \left[ \frac{\sin x \left( \frac{\pi - n\pi}{8} \right)}{\frac{\pi}{8} - \frac{n\pi}{8}} - \frac{\sin x \left( \frac{\pi + n\pi}{8} \right)}{\frac{\pi}{8} + \frac{n\pi}{8}} \right]_0^8$$

$$= 100 \left[ \frac{\sin(\pi - n\pi)}{(\pi - n\pi)} - \frac{\sin(\pi + n\pi)}{(\pi + n\pi)} \right]$$

Q7b)

Here, the Laplace eq<sup>n</sup> in polar form,

$$r^2 \frac{\partial^2 u}{\partial r^2} + r \frac{\partial u}{\partial r} + \frac{\partial^2 u}{\partial \theta^2} = 0$$

with boundary conditions

$$u(r, \theta) = 0^\circ \quad \text{in } 0 \leq r \leq 10$$

$$u(r, \pi) = 0^\circ \quad \text{in } 0 \leq r \leq 10$$